Is $\begin{pmatrix}2\\3\\1\end{pmatrix}$ a linear comb of $\begin{pmatrix}-1\\0\\1\end{pmatrix}$ and $\begin{pmatrix}1\\2\\1\end{pmatrix}$?

Elimination on $\begin{pmatrix}-1 & 1 & 2\\0 & 2 & 3\\1 & 1 & 1\end{pmatrix}$, If last col is free then yes
If last col is pivot then no.

Also can be posed as: Does $\begin{pmatrix}-1 & 1\\0 & 2\\1 & 1\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}2\\3\\1\end{pmatrix}$ have solutions?

$$\begin{array}{ccc} c1 & c2 & c3 \end{array}$$
$$\begin{pmatrix}-1 & 1 & 1\\0 & 2 & 4\\1 & 1 & 3\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\0 & 0 & 1\\0 & 1 & 0\end{pmatrix} = \begin{pmatrix}-1 & \boxed{1} & 1\\0 & \boxed{4} & 2\\1 & \boxed{3} & 1\end{pmatrix}$$

$\underline{c3 = c1 + 2c2}$. $\quad 3\times\underline{3}$ $\quad\quad \hookrightarrow \frac{1}{2}c2 - \frac{1}{2}c1$.

$3\times\underline{3}$

$$\begin{array}{ccc} c1 & c2 & c3 \end{array}$$
$$\begin{pmatrix}-1 & 1 & 1\\0 & 2 & 4\\1 & 1 & 3\end{pmatrix}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

$$\underline{c1} + \underline{2c2} - \underline{c3} = 0.$$

$$\begin{aligned}(A+B)^2 &= (A+B)(A+B)\\ &= A(A+B) + B(A+B)\\ \checkmark &= A^2 + AB + BA + B^2\\ &= \cancel{A^2 + 2AB + B^2}\end{aligned}$$

not true in general.

Find a matrix E such that for all A

$$A = \begin{pmatrix} - & a_1^T & - \\ - & a_2^T & - \\ - & a_3^T & - \end{pmatrix}\begin{array}{c} r_1\\ r_2\\ r_3\end{array}$$

(i) row 2 of EA = 2 r1 of A + r2 of A

(ii) row 1 & row 3 of EA = r1 of A and r3 of A resp.

$$E = \begin{pmatrix}1 & 0 & 0\\2 & 1 & 0\\0 & 0 & 1\end{pmatrix}$$

$$\begin{array}{ccccc} & & f & f & \\ \end{array}$$

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$
$$\begin{array}{ccccc} & & f & f & \end{array}$$

$$M = \begin{pmatrix} I_m & A_{m\times n} \\ \underline{B}_{n\times m} & I_n \end{pmatrix}$$

$I_m : m \times m$

$I_n : n \times n.$

$$\begin{bmatrix} \begin{pmatrix} 1 & & & 0 \\ 0 & 1 & & \vdots \\ 0 & & \ddots & 0 \\ 0 & & & 1 \end{pmatrix} \to I_m \\ \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & & & b_{2m} \\ \vdots & & & \vdots \\ b_{n1} & & & b_{n,m} \end{pmatrix} \end{bmatrix}$$

$$\begin{matrix} r_1 \\ r_1 \end{matrix} \begin{pmatrix} I_m & A_{m\times n} \\ \underline{B}_{n\times m} & I_n \end{pmatrix} \xrightarrow{r_2 \leftarrow r_2 - B_{n\times m} r_1.}$$

$$\downarrow$$

$$\begin{pmatrix} I_m & A_{m\times n} \\ \underline{0}_{n\times m} & I - BA \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \qquad \underline{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \qquad \underline{w} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

If A is square and you have a pivot in each col, rref (A) $= I$

$\therefore$ there is a matrix B so that

$$BA = I$$

Let C be a tall matrix with pivot in each col. Then

$$\text{rref}\left(\begin{bmatrix} n \times p \end{bmatrix} \right) = \begin{pmatrix} I_{p\times p} \\ 0_{(n-p)\times p} \end{pmatrix}$$

$$\therefore \quad \underset{n\times n}{D} C = \begin{pmatrix} I_{p\times p} \\ 0_{(n-p)\times p} \end{pmatrix}$$

Just look at the first p rows of D: $D_{p\times n}$. (remaining rows of D: $\underbrace{D'}_{\text{row } p+1 \to n}$.

$$D_{p\times n} C_{n\times p} = I_{p\times p}. \qquad D'C = \underline{0}.$$

$$C_{n\times p} D_{p\times n} \neq I_{n\times n}.$$

Kalman Filter

$$\underline{x} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \qquad \underline{y} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \qquad \underline{x}^T \underline{y} = 1(-1) + 2(1) + 1(-1) = 0 = y^T x$$

$$\underline{x}\,\underline{y}^T =$$

$$I + \underline{x} y^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 1 & -1 \\ -2 & 2 & -2 \\ -1 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & -1 \\ -2 & 3 & -2 \\ -1 & 1 & 0 \end{pmatrix}.$$

$$1 + \underline{y}^T \underline{x} = 1 + 0 = 1$$